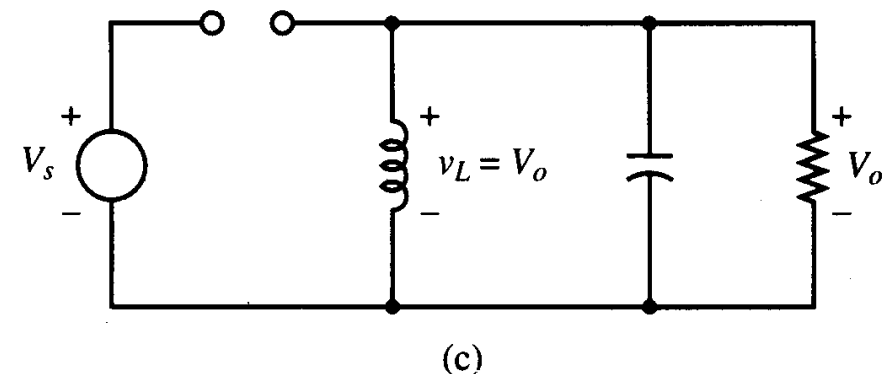
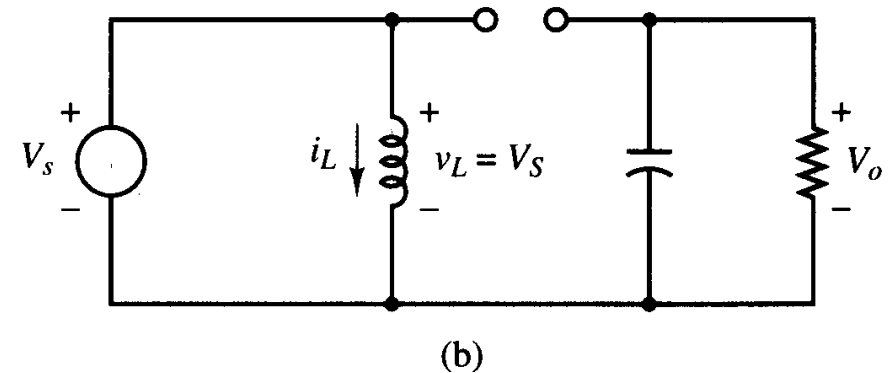
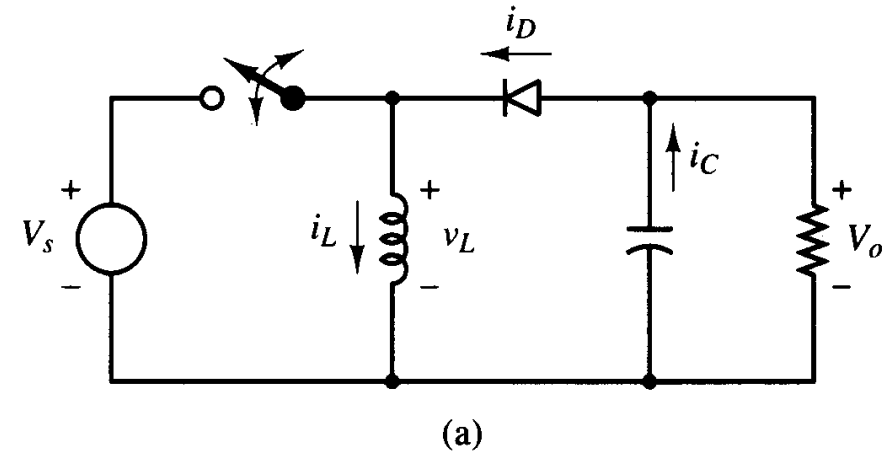


DC-DC CONVERTERS – BUCK-BOOST

BUCK-BOOST-1 BUCK-BOOST CONVERTER

The *buck-boost converter* again uses essentially the same set of components as the buck and boost converters, arranged differently as shown in Figure 2.1. Called ‘buck-boost’ since V_o can be **both** less than and greater than V_s .

Analysis is based on the normal **steady state** properties.

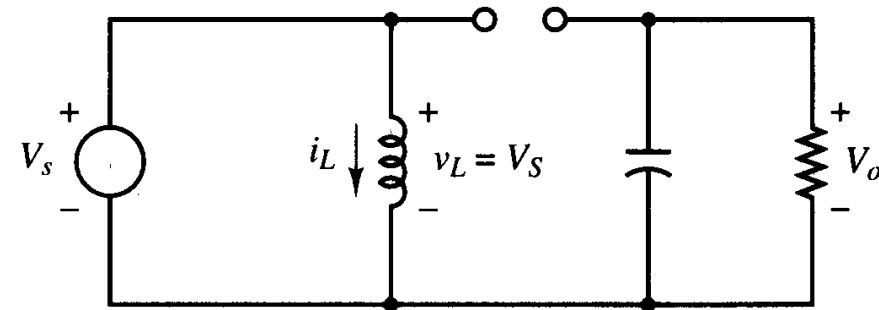


BUCK-BOOST-1.1 ANALYSIS

Switch Closed

When the switch is closed, the diode is reverse biased which results in,

$$v_L = L \frac{di_L}{dt} = V_S \quad \text{or} \quad \frac{di_L}{dt} = \frac{V_S}{L}$$



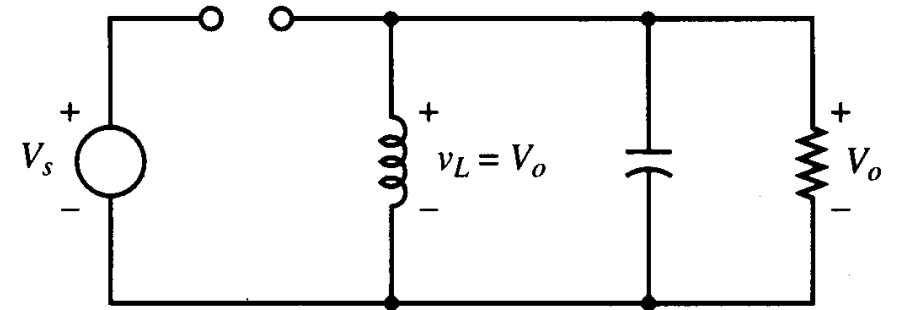
Since V_s **and** L are constants, then the time derivative of the inductor current is a constant. That means, the inductor current must be **constantly ramping up** while the switch is closed, where

$$\Delta i_{L, \text{closed}} = \frac{V_s DT}{L}$$

This is identical to the **boost** converter.

Switch Open

When the switch is open, the inductor voltage reverses to keep the current flowing in the same direction, which forward biases the diode. In this situation the inductor is effectively in parallel with the load.



So (using the sign convention in the figures),

$$v_L = L \frac{di_L}{dt} = V_o$$

Again the derivative of the current is a constant, so the current from the inductor is **constantly ramping down** while the switch is open, where

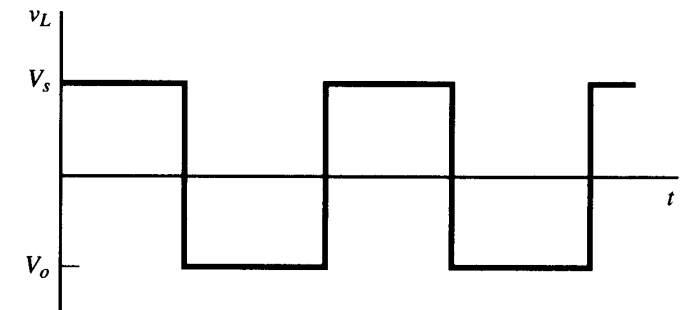
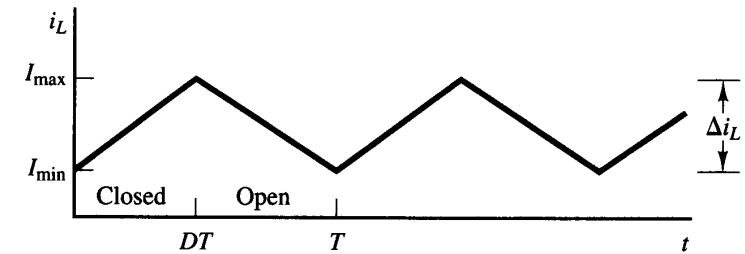
$$\Delta i_{L,open} = \left(\frac{V_o}{L} \right) (1 - D) T$$

Which is identical to the **buck** converter.

Since the average fluctuation in inductor current must equal zero (steady-state), then it turns out that

$$V_O = -V_s \left(\frac{D}{1-D} \right)$$

And since D lies somewhere between 0 to 1, the output voltage can be **greater** than or **less** than the source voltage, but always of **opposite polarity**.



Continuous Current Design Consideration

To keep the inductor current continuous, its minimum current must not drop to zero. Following the same analysis we have done before for the buck and boost converters, it can be shown that at the point where the current just touches on zero,

$$L > \frac{V_s(1-D)D}{2f_s I_o}$$

Output Voltage Ripple

If there is current flowing in the capacitor, then there must be an associated voltage variation. By consideration of the same voltage-charge relationship for capacitors as for the boost converter, it turns out that

$$\Delta V_o = \frac{I_o D}{f_s C}$$

Which is an identical expression to the **boost** converter.

NONIDEAL-1 NONIDEAL COMPONENT EFFECTS

An assumption made at the beginning of the DC-DC converter analyses was that all the components were ideal. As we will see, incorporating more realistic component behaviour can result in sometimes significant altering of circuit parameters (especially the output voltage). That means these possible effects need to be considered in any practical design.

NONIDEAL-1.1 SWITCH VOLTAGE DROPS AND LOSSESS

Obviously any real DC-DC converter will use switches that are not ideal. Real switches have a non-zero on impedance (and therefore some voltage drop) and take a non-zero time to switch on and off.

A good example of how a non-zero switch voltage affects output voltage can be seen with the buck converter. If the switch voltage is given as V_Q and the diode voltage is given as V_D then the modified output voltage is,

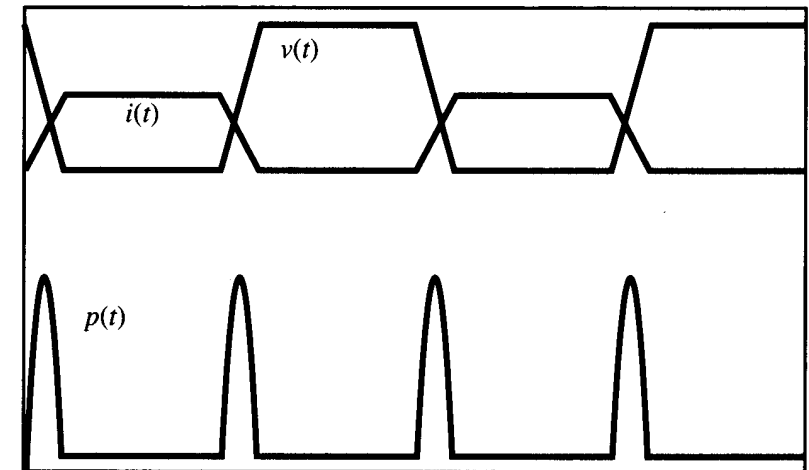
$$V_O = V_S D - V_Q D - V_D (1 - D)$$

Which is less than the ideal output voltage.

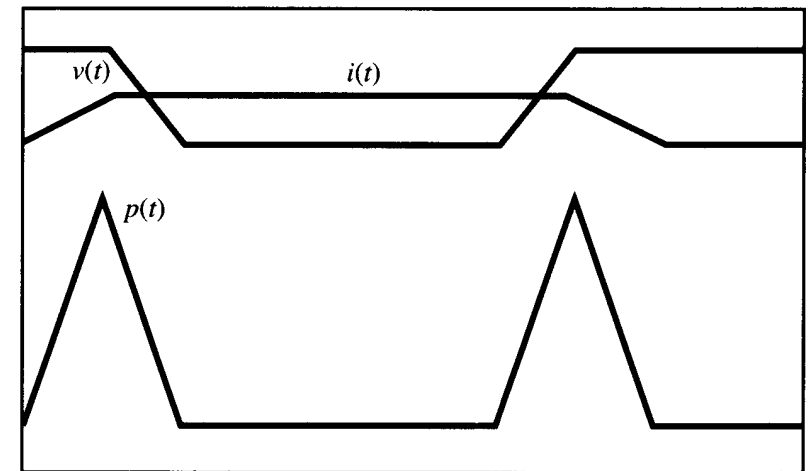
For low voltage and/or high current converters, the switch voltages can become a significant percentage of the output voltage and must be included in design calculations.

Switching losses occur whenever both the current and voltage associated with the switch are non-zero. Voltage **and** current at the same time means **power**, and since the switch itself is not a source of power, that power must be dissipated as heat from the switch. This situation is depicted in Figure 1.1. Two switching possibilities are shown, with voltage and current switching simultaneously (Figure 1.1 (a) – the best situation you could hope for)), and switching alternatively (Figure 1.1 (b) – the worst situation for switching). In practice, most switches will fall in between these situations, but on average slightly to the worst side.

Figure 1.1 (a) Practical switching characteristics and power (losses) for optimal switching situation. (b) Practical switching characteristics and losses for worst-case switching situation.



(a)



(b)

NONIDEAL-1.2 CAPACITOR SERIES RESISTANCE AND INDUCTANCE

All physical capacitors have an inherent (or equivalent) series resistance r_C **and** series inductance l_C . These are known as the capacitor **ESR** and **ESL** respectively. Each of these factors has an effect on the output voltage.

As we saw with all the converter analysis, there were appreciable filter capacitor currents (especially peak currents). By Ohm's law the capacitor ESR results in an associated voltage drop across the capacitor, where

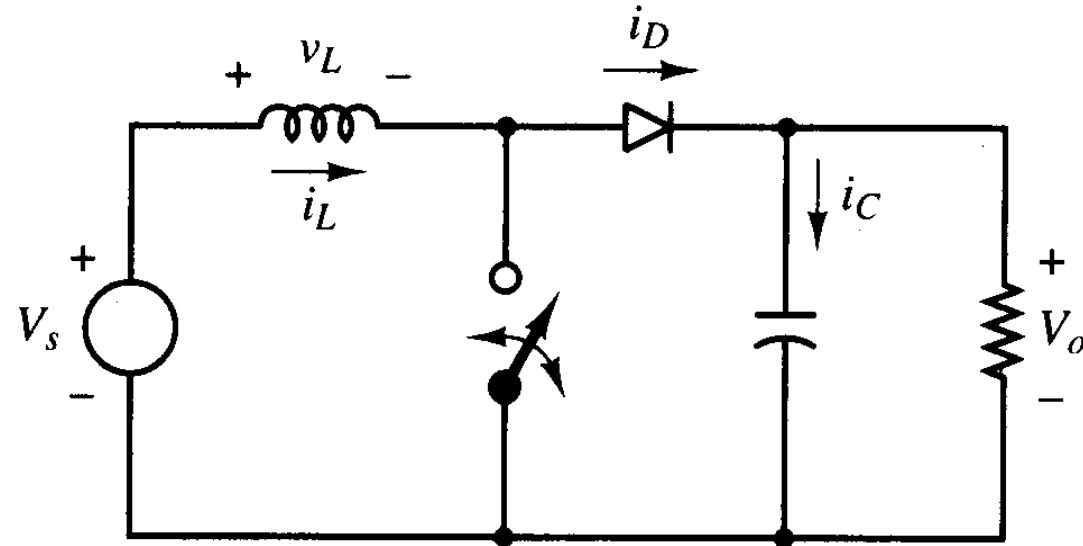
$$\Delta V_{O,ESR} = \Delta i_C r_C$$

The voltage drop appears as added ripple on the output voltage. Since there can be quite large peak filter capacitor currents, the ESR can generate significant output voltage ripple.

Series inductance represents quite a different problem. The equivalent series CLR circuit representing the capacitor will resonate at a certain frequency. If this frequency is excited by the switching operation then it is possible for the output voltage to become severely attenuated due to the drop in overall real capacitor impedance at resonance. The effect is usually only observed in the high hundreds of kilohertz switching frequencies.

NONIDEAL-1.3 INDUCTOR RESISTANCE

Since inductors are made from coils of wire, they must have some finite amount of resistance r_L . The effect of this equivalent series resistance is well illustrated in the effect on output voltage for the boost converter.



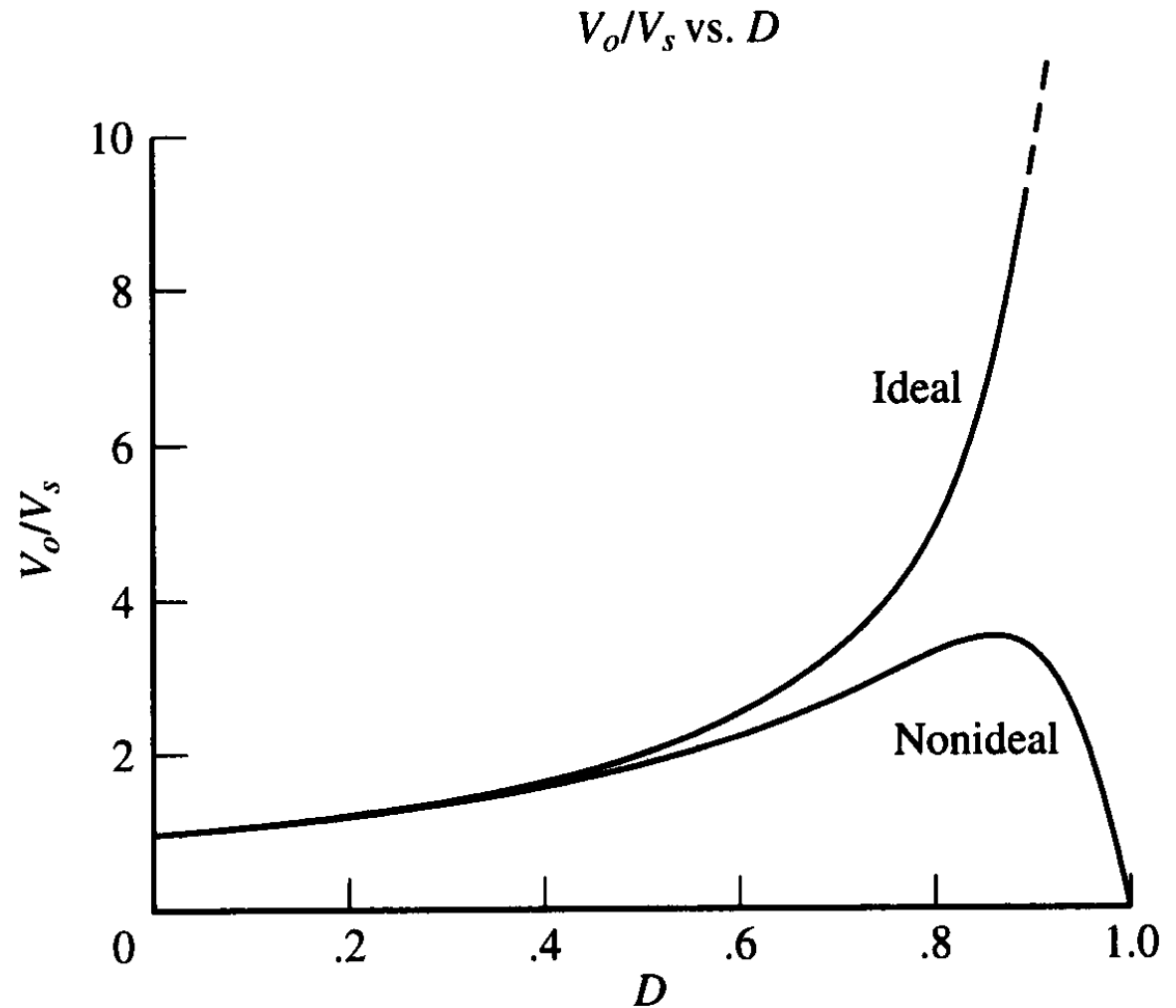
If we set the inductor to be very large so that its current is essentially constant and accept that the only output power loss is due to the inductor resistance, such that

$$P_S = P_O + P_{r_L}$$

Then it can be shown that,

$$V_o = \frac{V_s}{1-D} \left(\frac{1}{1 + \frac{r_L}{R(1-D)^2}} \right)$$

The implications of the inductor ESR are twofold. Firstly, as r_L increases, the output voltage decreases (not usually too much of a problem since r_L is often quite small). Secondly, and more importantly, as the duty ratio increases (to try and increase output voltage) then beyond a certain point the output voltage will actually start to decrease rather than increase as shown in this figure.



NONIDEAL-1.3 TRANSIENTS

Another feature of practical converter operation and design we have not looked at are transients. These occur as start-up transients and fast input voltage and/or load change transients. Transients can produce voltages and currents that might damage components that would otherwise be fine for steady state conditions.

Analysis of transient behaviour of PE circuits is a little beyond the scope of this course. Just be aware that they exist and may prove to be a problem. Transient effects at start-up can be significantly reduced through what is known as soft-starting where the duty ratio is slowly increased from zero to its final value over a period of 10's or 100's of milliseconds.